

LAB # 9

Abstract Data Types

Lab Tasks:-

- Considering a bank interface, it will have different concrete classes for receiving bill payments, opening of new accounts and contacting those loan takers whose limits of extension have been expired. Now, develop this scenario on the described ADT and decide which ADT should be used for each concrete class, also implement the whole scenario in multiple java classes.

Code:-

```

1 public interface BankOperation {
2     void add(String item);
3     void process();
4 }

1 import java.util.LinkedList;
2 import java.util.Queue;
3
4 public class BillPayment implements BankOperation {
5     Queue<String> queue = new LinkedList<>();
6
7     public void add(String bill) {
8         queue.offer(bill);
9     }
10
11    public void process() {
12        System.out.println("Processing Bill Payments:");
13        while (!queue.isEmpty()) {
14            System.out.println("Paid: " + queue.poll());
15        }
16    }
17 }

1 import java.util.ArrayList;
2 import java.util.List;
3
4 public class NewAccountOpening implements BankOperation {
5     List<String> accounts = new ArrayList<>();
6
7     public void add(String accName) {
8         accounts.add(accName);
9     }
10
11    public void process() {
12        System.out.println("New Accounts Created:");
13        for (String acc : accounts) {
14            System.out.println("Account created for: " + acc);
15        }
16    }
17 }
  
```

BankOperation.java
BillPayment.java
NewAccountOpening.java

```

1      import java.util.Stack;
2
3      public class LoanDefaulters implements BankOperation {  no usages
4          Stack<String> stack = new Stack<>();  3 usages
5
6      public void add(String defaulter) {  no usages
7          stack.push(defaulter);
8      }
9
10     public void process() {  no usages
11         System.out.println("Contacting Loan Defaulters:");
12         while (!stack.isEmpty()) {
13             System.out.println("Contacted: " + stack.pop());
14         }
15     }
16 }

```

BankOperation.java
BillPayment.java
NewAccountOpening.java

```

1  public class Main2 {
2      public static void main(String[] args) {
3
4          BankOperation bills = new BillPayment();
5          bills.add("Electricity Bill");
6          bills.add("Water Bill");
7
8          BankOperation accounts = new NewAccountOpening();
9          accounts.add("Rayyan");
10         accounts.add("Wania");
11
12         BankOperation loanDefaulters = new LoanDefaulters();
13         loanDefaulters.add("Sharjeel");
14         loanDefaulters.add("Faria");
15
16         bills.process();
17         System.out.println();
18
19         accounts.process();
20         System.out.println();
21
22         loanDefaulters.process();
23     }
24 }

```

Output:-

```
C:\Users\lab2\.jdk\openjdk-25\bin
Processing Bill Payments:
Paid: Electricity Bill
Paid: Water Bill

New Accounts Created:
Account created for: Rayyan
Account created for: Wania

Contacting Loan Defaulters:
Contacted: Faria
Contacted: Sharjeel

Process finished with exit code 0
```